Manpower Roster Optimisation

A campus-placement assignment for the NCR distribution centre. Across a **7-day week**, a Forward shipment stream is processed through two sequential stages — **Primary then Secondary**, with a one-hour gap between them. Design an optimised **monthly** manpower roster across FTC, Flex and PT that processes the entire volume at the **lowest monthly cost**. You may use **any AI tool** to solve it, and must submit your full prompt history.

OBJECTIVE Lowest monthly cost · entire weekly volume processed · Primary → Secondary sequence

WEEKLY VOLUME	DAYS	PEAK DAY	STAGES
1,98,000	**7**	**36,000**	**2**
Mon–Sun, Forward	Monday = Day 1	Friday	Primary → Secondary

01 Daily Shipment Volume

The week starts on **Monday (Day 1)**. Daily incoming Forward volumes:

DAY	INCOMING VOLUME	DAY	INCOMING VOLUME
Monday	30,000	Friday	36,000
Tuesday	35,000	Saturday	35,000
Wednesday	25,000	Sunday	10,000
Thursday	27,000		
		Weekly total	**1,98,000**

This 7-day pattern **repeats every week** of the month (≈ 4.33 weeks), giving a monthly volume of about **8,57,000** shipments. The roster is prepared on a **monthly basis** — a day-on-day attendance plan across the full month.

02 Hourly Volume Distribution

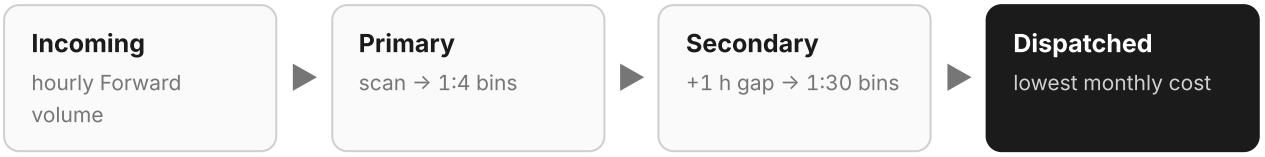
The **hourly contribution percentages are unchanged** from the earlier model — only the daily total changes. Each day's hourly volume = **day total × hourly %**. Apply the same profile to every day.

HOUR	%	MON (EX.)	HOUR	%	MON (EX.)
00:00	2.73	818	12:00	4.55	1,364
01:00	2.73	818	13:00	3.64	1,091
02:00	1.82	545	14:00	3.64	1,091

HOUR	%	MON (EX.)	HOUR	%	MON (EX.)
03:00	1.82	545	15:00	4.55	1,364
04:00	0.91	273	16:00	5.45	1,636
05:00	0.91	273	17:00	6.36	1,909
06:00	1.82	545	18:00	7.27	2,182
07:00	2.73	818	19:00	8.18	2,455
08:00	3.64	1,091	20:00	7.27	2,182
09:00	4.55	1,364	21:00	6.36	1,909
10:00	5.45	1,636	22:00	4.55	1,364
11:00	5.45	1,636	23:00	3.64	1,091

"Mon (ex.)" shows Monday's 30,000 split by the profile; repeat the same percentages for every other day using that day's total.

03 Shipment Processing Flow



Primary. Every shipment is first scanned, then sorted into four directional bins — **North, South, East, West** — a **1:4 sort**. This is broad-level sorting, so per-person capacity is **higher**.

Secondary. Shipments then move to Secondary **one hour after** Primary, in sequence, and are sorted into **30 bins** by destination hub area — a **1:30 sort**. This much finer sorting means per-person capacity is **lower** than Primary.

The model must respect the sequence (Primary before Secondary), the **one-hour gap**, and the **capacity difference** between the two stages.

04 Manpower Categories and Parameters

CATEGORY	DEFINITION
FTC — Full-Time Contractual	Full-time manpower deployed on a regular basis, with defined working hours and monthly cost. Forms the **base** of the roster.
Flex	Engaged whenever additional capacity is required, depending on shipment volume.
PT — Part-Time	Available for shorter working hours and specific operational requirements.

CATEGORY	WORKING HOURS	PRIMARY CAP / DAY	SECONDARY CAP / DAY	COST
FTC	8 h	1,000	500	**₹20,000 / month**
Flex	8 h	800	400	**₹700 / day**
PT	4 h	600	300	**₹12,000 / month**

NOTE Capacity is shipments processed per person per day at the stage the employee is assigned. **Weekly offs apply to FTC only;** Flex and PT do not have weekly offs. **Flex and PT may each be deployed up to a maximum of 10% of total manpower**, on particular (peak) days; FTC forms the remaining base (at least 80%). The 7-day pattern repeats each week; build the roster and cost at **monthly level** (FTC and PT monthly; Flex per deployment day).

05 Roster Design and Mandatory Outputs

Develop a manpower roster covering the entire shipment volume, prepared on a **monthly basis** (the 7-day pattern repeating each week), with FTC as the base (**≥ 80% of manpower**) and Flex / PT **each capped at 10%**, used on particular peak days to manage fluctuations and minimise cost. The roster is prepared at **employee level** — each individual employee, their category, stage and shift, and their **day-on-day attendance across the month**. All candidates must submit the **same four outputs in the same formats** (Section 06) for objective comparison:

1. **Manpower Roster** — **employee-level, day-on-day present / absent across the month**, with each employee's category, stage (Primary / Secondary) and shift hours.
2. **Hourly Volume Planning** — incoming volume, Primary processed volume, Secondary processed volume and pending volume.
3. **Monthly Cost Calculation** — category-wise cost, deployment days, total monthly cost and cost optimisation.
4. **AI Prompt History** — the complete AI prompt conversation exported as a PDF (Ctrl + P → Save as PDF).

AI TOOLS Candidates may use **any AI tool or module** (ChatGPT, Claude, Gemini, Copilot or any other). The complete prompt history must be submitted.

06 Standardised Output Formats

Fill these exact formats so submissions are directly comparable. Dashes mark cells to be completed.

Format 1 — Manpower Roster (employee-level, monthly day-on-day present / absent)

A **monthly** attendance roster: one row per employee, marked Present / Off for **every day of the month (D1–D30)**. Each employee has a fixed category, stage and shift; the 7-day volume pattern repeats weekly. Repeat for every employee.

EMP ID	CAT.	STAGE	SHIFT	D1	D2	D3	D4	D5	D6	D7	...	D30	MO. DAYS
E001	FTC	Primary	09–17	P	P	P	P	P	O	P	...	P	26
E002	FTC	Secondary	10–18	P	O	P	P	P	P	P	...	P	26

EMP ID	CAT.	STAGE	SHIFT	D1	D2	D3	D4	D5	D6	D7	...	D30	MO. DAYS
P010	PT	Primary	18–22	P	P	P	P	P	P	P	...	P	30
F004	Flex	Secondary	17–21	–	P	–	–	P	P	–	...	–	12
...	...	...	...	...	...	...	...	...	...	...	...	...	...

P = present · O = weekly off (FTC only) · – = not engaged (Flex). "Mo. days" = present days in the month, carried into the cost calculation (Format 3).

Format 2 — Hourly Volume Planning

One row per Day × Hour. "Pending" = volume awaiting the next stage at the end of the hour.

DAY	HOUR	INCOMING	PRIMARY PROCESSED	SECONDARY PROCESSED	PENDING
Monday	00:00	818	–	–	–
Monday	01:00	818	–	–	–
...	...	...	...	...	...
Sunday	23:00	–	–	–	–

Format 3 — Monthly Cost Calculation

CATEGORY	HEADCOUNT	DEPLOYMENT DAYS	RATE	MONTHLY COST
FTC	–	–	₹20,000 / month	–
Flex	–	–	₹700 / day	–
PT	–	–	₹12,000 / month	–
Total	**–**			**–**

Add a short **cost-optimisation note**: the chosen FTC / Flex / PT mix, and the saving versus an all-FTC baseline.

Format 4 — AI Prompt History

Export the complete AI conversation (any tool) as a PDF — press **Ctrl + P** and choose **Save as PDF** — showing your approach, iterations and problem-solving.

07 Final Objective and Evaluation

Design an optimised monthly manpower roster that processes the entire incoming volume, follows the Primary → Secondary sequence, accounts for the one-hour gap, optimises FTC / Flex / PT deployment, and minimises the overall monthly manpower cost.

Evaluation considers **accuracy of calculations, operational understanding, roster optimisation, cost efficiency and the quality of the problem-solving approach** — and whether the four outputs follow the standard formats.